
SENTIMENT & POSITIONING

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Abstract

Sentiment is the most discussed and least disciplined input in discretionary trading. This paper treats it as a set of *measurable, testable quantities* rather than a feeling. We separate three families — survey-based (AAII, Investors Intelligence, fund-manager surveys), market-based (put/call, skew, the VIX complex), and flow-based (fund flows) — and are explicit about which are free and which are paywalled. We decode the CFTC Commitments-of-Traders report, confront the commercials-versus-speculators frame with the evidence that it rarely predicts, and argue that options positioning is the cleanest real-time gauge. We build a three-family composite on free data and show its central defect: two of its three inputs measure the same fear axis, so equal weighting double-counts it. We then pre-register a contrarian test — do extreme readings reverse? — and report, honestly, that the VIX fear decile is followed by materially higher forward returns out-of-sample while the *complacency* half of the claim fails, and that the composite is too thin to test. We close with text sentiment, crowding and fragility (Feb-2018, 2021), and the limits that make most sentiment "edges" survivorship. Every figure is reproducible on free data.

Keywords — Investor sentiment, positioning, Commitments of Traders, put/call ratio, VIX, skew, dealer positioning, options flow, composite indicator, multicollinearity, contrarian, pre-registration, out-of-sample, variance risk premium, crowding, deleveraging, text sentiment, look-ahead bias

1 Introduction

1.1 Motivation and thesis

"Sentiment is bullish." "Everyone is offside." "The crowd is euphoric." These sentences feel like analysis and function like horoscopes: they are true after the fact, whichever way the market went, because none of them names a number, a threshold, or a horizon. The purpose of this paper is to refuse that comfort. Sentiment and positioning are not moods to be intuited; they are *quantities that can be measured, normalized, combined, and tested* — and the moment you insist on measuring them, most of the folklore collapses.

The thesis is a chain of three claims, each stronger than the last. First, *sentiment is measurable*: for every vague affect there is at least one free, dated series that operationalizes it, and choosing the series is the first honest decision. Second, *the measurements are redundant*: the market-based gauges — put/call, skew, the VIX — mostly track one underlying fear factor, so a naïve "composite" of five indicators is often one indicator counted five times. Third, and most consequentially, *the tradable content of extreme sentiment is real but asymmetric and regime-dependent*: buying extreme fear has paid, out-of-sample, in the modern sample we test; selling extreme complacency has not. The popular contrarian slogan — "extremes reverse" — is half right, and the wrong half is the one most people act on.

None of this is anti-sentiment. It is a demand that sentiment be held to the same standard as any other signal in this library: a pre-registered hypothesis, an out-of-sample split, an effect size with a confidence interval, and an explicit account of where it failed. A sentiment indicator that has never been tested this way is not a signal; it is a story that happens to be quantified.

1.2 Scope, prerequisites, and intended reader

This research paper sits in the Trading Library's *Macro & Sentiment* track. It assumes the macro-regimes paper — the reader should be comfortable with the idea that returns are conditioned by a slowly moving state variable — and the backtesting paper: pre-registration, the researcher-degrees-of-freedom problem, and the operational meaning of opening an out-of-sample set once. For the positioning material it leans on the options-microstructure track: the mechanics of *dealer positioning* and *gamma* from the dealer-flows-and-GEX paper and the *deleveraging cascade* from the flow-landscape paper are used here as consequences, not re-derived. It assumes a reader who can pull a CSV and run a short Python script from free data; it assumes no proprietary feed and no options-pricing theory beyond the intuition that a put costs more when people are scared.

After reading this paper, a reader can:

- name, for any sentiment claim, which of the three families it belongs to, whether the underlying series is free or paywalled, and which bias it is most exposed to;
- read a Commitments-of-Traders report without falling for the commercials-versus-speculators morality tale, and state a percentile rather than a direction;
- build a normalized, reproducible sentiment composite on free data — and then run the correlation matrix that tells them how much of it is one signal in a trench coat;
- pre-register a contrarian test with a null, a directional alternative, a fixed split, and a decision rule, and read the resulting effect sizes with confidence intervals instead of anecdotes;
- distinguish the part of "buy the fear" that is a *sentiment* effect from the part that is a *variance risk premium* — a distinction most contrarian write-ups quietly skip;
- describe a minimal, leak-free text-sentiment pipeline and name the look-ahead traps unique to text data;
- and connect a positioning extreme to the flow trigger that turns it into the kind of deleveraging cascade the flow-landscape paper traces — sentiment as the fuel, flows as the match.

1.3 Data and reproducibility

All figures and numbers use freely available data: the CBOE VIX and (discontinued) equity put/call files, the CFTC Commitments-of-Traders history, and Yahoo daily S&P 500 closes. Each figure caches its downloads under `figures/data/` and is reproducible from the cited script. Where the argument draws on a private trade list or a proprietary options feed, that fact is stated and the data is not redistributed; every published number is recomputed on free data instead. *This is a research and educational document, not investment advice.*

Because this paper is about causation in a reflexive system, one rule binds every "X leads to Y" sentence: it must be accompanied either by a historical counterexample or by an explicit statement of a period in which it did not hold. A causal claim with no named exception is treated as unproven.

1.4 Organization of this paper

The paper is complete; the sections below are all present, and the arc is deliberate. Section 2 fixes what "sentiment" *means* operationally — the three families (survey, market, flow), the three biases they carry, and an explicit map of what is free versus paywalled. Section 3 turns to *positioning*, the harder-nosed cousin of sentiment: the Commitments-of-Traders report decoded (with the current speculator percentile computed live), ETF and fund-flow proxies, and the argument that options positioning is the cleanest real-time gauge. Section 4 builds a reproducible *composite* on free data and immediately runs the correlation matrix that exposes its

multicollinearity. Section 5 is the empirical heart: a pre-registered *contrarian* test with an out-of-sample split, reporting effect sizes and confidence intervals for the single cleanest series (the VIX) and for the composite, and confronting the hard truth about when the crowd is right. Section 6 treats *news and text* sentiment — the dictionary-versus-model choice, the leakage traps unique to text, and a minimal reproducible pipeline described end to end. Section 7 connects positioning extremes to *crowding and fragility*, using February 2018 and the 2021 meme episode as case studies of sentiment-as-fuel and flows-as-trigger. Section 8 states the *limits* — noise, revision, regime-dependence, the survivorship of indicators that "worked" — and imports the minimum-cell discipline from the Quant track. Section 9 gives annotated references and the changelog.

2 What "sentiment" means operationally

2.1 Three families, and why the distinction is not cosmetic

Ask ten traders to define sentiment and you will get ten metaphors. Ask them to *measure* it and the metaphors collapse into three families, distinguished by where the number comes from:

Family	What it measures	Canonical free series	Canonical paywalled series
Survey	What people <i>say</i> they feel or will do	AAII bull/bear (weekly, free)	Investors Intelligence, BofA Fund Manager Survey
Market	What people <i>pay</i> to express a view	VIX, VIX term structure, equity put/call (free, some discontinued)	Live skew surfaces, dealer-gamma estimates
Flow	What people <i>actually did</i> with money	ETF flows via 13F/creation data (lagged, free-ish)	Real-time fund-flow trackers (EPFR, Lipper)

The families are ordered, roughly, by how much skin is in the number. A survey respondent risks nothing by ticking "bearish"; an option buyer has posted premium; a fund that moved a billion dollars has already paid the price impact. This ordering matters because *the biases differ by family* (Section 2.5), and a composite that mixes families without accounting for it inherits all three biases at once. It is also why the same word — "bullish" — can be simultaneously true in one family and false in another, and why the honest first question about any sentiment claim is not "is it bullish?" but "*in which family, measured how, and free or paywalled?*"

2.2 Survey-based sentiment

Surveys ask a sample of participants to state a view and tabulate the answers. The best-known free series is the *American Association of Individual Investors* (AAII) weekly poll, which since 1987 has reported the share of respondents who are bullish, bearish, or neutral on the next six

months. The *Investors Intelligence* advisor-sentiment survey (bull/bear among newsletter writers) and the monthly *BofA Global Fund Manager Survey* (cash levels, overweight/underweight positioning of large allocators) are widely cited but sit behind paywalls or subscription reports; their headline numbers leak into the press with a lag, which is exactly the kind of look-ahead-prone, hard-to-timestamp series a reproducible study should avoid.

A word on construction, because the folklore glosses it. AAI reports three shares — bullish, bearish, neutral — that sum to one; the quantity practitioners actually watch is the *bull-bear spread* (bullish minus bearish), which centres near a small positive number and oscillates. Investors Intelligence reports a bull share and a bear share among newsletter writers whose ratio (bulls/bears) is the classic advisor-sentiment gauge, with readings above ~ 3 flagged as complacent and below ~ 1 as fearful in the practitioner tradition. The two are often blended into a *smart-money-versus-dumb-money* narrative: newsletter writers and individual investors as the "dumb" retail crowd to be faded, institutions and commercial hedgers as the "smart" money to be followed. The narrative is memorable and, as Section 3.2 will show for its positioning cousin, poorly supported by out-of-sample evidence — the labels encode a prejudice (retail is wrong, institutions are right) that the data does not consistently reward. Treat the smart/dumb framing as a mnemonic for *who is crowded*, not as an identity of who is correct.

Two facts about surveys are underappreciated. First, *the level is nearly useless; the extreme and the change carry what little signal there is*. AAI bullishness spends most of its life in a wide, noisy band; the academic evidence that survey sentiment forecasts returns is concentrated in the tails and is statistically weak even there — Brown and Cliff (2005) find that sentiment relates to *valuation* and subsequent multi-year returns more than to the next week (Brown & Cliff 2005). Second, *surveys are cheap talk*: because respondents bear no cost, the number is contaminated by response bias, framing, and the simple fact that people report the mood of the recent tape rather than a forecast. The AAI series is, empirically, strongly *contemporaneously* correlated with trailing returns — it is closer to a lagging sentiment thermometer than a leading indicator. This is the first family and, tellingly, the one we do *not* build a figure around: the free survey series are the weakest link, and pretending otherwise would violate the paper's own standard.

A causal claim with its exception, stated plainly: *extreme AAI bearishness has often preceded rebounds* — but it did **not** hold as a timing tool through 2008, when bearishness reached record extremes in January and the market fell another $\sim 45\%$ into March 2009 before the "contrarian buy" paid. The extreme was correct about *eventual* mean-reversion and catastrophically wrong about *when*. Survey extremes are a statement about multi-year valuation, not a trade date.

2.3 Market-based sentiment

Market-based gauges read sentiment off prices people actually pay, and they dominate the practitioner toolkit for one reason: they are timestamped, continuous, and costly to fake. Three matter.

The *equity put/call ratio* is the volume of equity puts traded divided by the volume of equity calls. High readings mean demand for downside protection (fear); low readings mean call-buying (greed or hedging unwind). The CBOE published a free daily equity put/call file from November 2006, and — a fact every reproducibility-minded researcher should internalize — **discontinued the free download in October 2019**. Figure 2 uses the full 2006-11-01 to 2019-10-04 history (3,253 sessions); anything after that requires a paid feed, so a "put/call strategy" backtested to the present on free data is quietly using a series that no longer exists for free.

Skew — the price of out-of-the-money puts relative to at-the-money or out-of-the-money calls — is the options market's asymmetry premium. Steep skew means puts are bid up relative to calls: crash insurance is expensive, which is usually read as fear, though it is as much a structural feature of equity-index options (persistent demand for protection from institutions) as a sentiment reading. Live skew surfaces are paywalled; the CBOE's SKEW index is free but famously noisy and hard to interpret as a directional signal.

The *VIX complex* is the workhorse. The VIX is the 30-day implied volatility of the S&P 500, free from CBOE back to 1990 and, via a proxy, to 1986. Whaley (2000) christened it "the investor fear gauge," and the name is apt: the VIX rises when demand for index protection rises (Whaley 2000). Two refinements make it a sentiment *system* rather than a single number. The *term structure* (VIX versus VIX3M or the VIX futures curve) distinguishes acute fear — a front-month spike above the back, i.e. *backwardation* — from a calm, upward-sloping *contango*; backwardation is a sharper, rarer fear signal than a high VIX level alone. And *VVIX*, the vol-of-vol, measures the price of VIX options and captures fear about fear. For a long-history, free, single sentiment series, the VIX is the least-bad choice, and it is the one we carry through the contrarian test in Section 5.

It is worth pausing on *why* the term structure adds information that the level does not. The VIX front month prices the next 30 days; VIX3M prices the next 90. In calm markets the curve slopes up (contango) because uncertainty compounds with horizon and because sellers of longer protection demand more premium — this is the resting state, and a high-but-contango VIX is "expensive insurance, no panic." When the front month jumps *above* the back (backwardation), the market is pricing acute near-term risk above chronic risk: this is the shape that accompanies genuine stress, and it is far rarer and far more informative than a merely elevated level. The ratio VIX/VIX3M is thus a cleaner fear switch than VIX alone, and it is free. VVIX adds a second dimension: it is the implied volatility of VIX *options*, so it prices the market's uncertainty about

volatility itself — fear about fear — and a VVIX spike without a VIX spike has historically flagged demand for convex tail hedges before the tail arrived. The honest caveat, again the recurring one: these refinements sharpen the *fear reading*, but every one of them is still contaminated by the risk premium described in Section 2.5, so a "backwardation signal" is part information and part the price of insurance in a genuinely dangerous moment.

Figure 2 shows the two market-based gauges side by side, and it makes the discontinuity concrete. The equity put/call ratio (top, 21-day moving average) marks its fear extremes (top 5%, ≥ 0.777) and complacency extremes (bottom 5%, ≤ 0.558), then simply *ends* in October 2019. The VIX (bottom, log scale, 1990-2026) marks 208 sessions closing above 40 and shows a latest reading of 17.71 — the 50.8th percentile of its entire history, i.e. dead average. The two gauges are positively correlated on their overlap (+0.545), which is a preview of Section 4's problem: they are not independent readings, they are two windows onto the same fear.

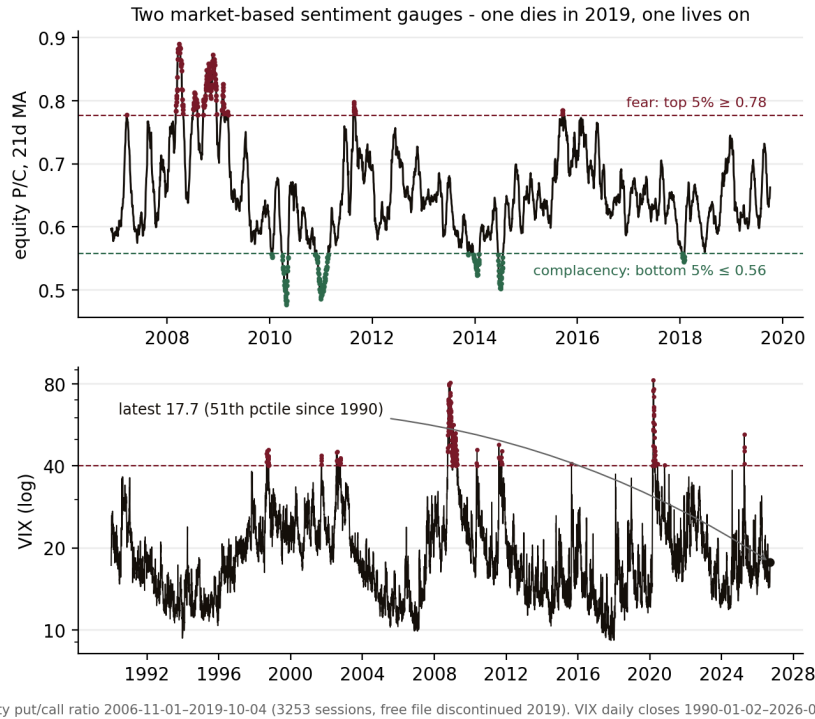


Figure 1. Two market-based sentiment gauges over their free histories. Top: the CBOE equity put/call ratio (21-day moving average, 2006-11-01 to 2019-10-04, 3,253 sessions) with its top-5% fear threshold (0.777) and bottom-5% complacency threshold (0.558) marked; the free file was discontinued in October 2019 and the series simply ends. Bottom: the VIX (log scale, 1990-2026, 9,274 sessions), 208 closes above 40 marked, latest 17.7 at the 50.8th historical percentile. The two gauges correlate +0.55 on their overlap. Reproduce with figures/fig_e3_putcall.py.

2.4 Flow-based sentiment

Flows measure what money actually did. The purest version is *fund flows*: net creations and redemptions in mutual funds and ETFs, which reveal whether the marginal dollar is entering or leaving an asset class. The appeal is that a flow is an accomplished act, not a stated intention — the highest rung on the skin-in-the-game ladder. The problem is data access and timing. High-frequency flow trackers (EPFR, Lipper) are paywalled; the free proxies — ETF shares outstanding from creation/redemption data, or 13F holdings — are either coarse or badly lagged, and 13F in particular is filed 45 days after quarter-end, which makes it useless for timing and dangerous for backtests that forget the lag.

Flow sentiment has one robust empirical regularity and one robust warning. The regularity: *retail fund flows chase performance* — money arrives after returns, not before — so aggregate equity-fund inflows are a *coincident-to-lagging* indicator of the mood, not a leading one (the performance-chasing literature is old and consistent). The warning is the exception that guts the naïve trade: "flows into equities are extreme, therefore fade" is *not* a reliable rule, because sustained inflows can accompany sustained bull markets for years (1995-1999; 2013-2019). Flows tell you the crowd is committed; they do not tell you the crowd is wrong, and the interval between "committed" and "wrong" has been measured in years, not weeks.

2.5 Three families, three biases

Each family carries a characteristic distortion, and knowing which one lets you discount the number correctly:

- **Surveys carry response bias and reflexivity.** Cheap talk tracks the recent tape; respondents report mood, not forecast, and the sample self-selects. The measured "sentiment" is partly just lagged returns wearing a survey's clothes.
- **Market gauges carry a risk-premium contamination.** A high VIX is *fear*, but it is also a *price*: index puts are structurally expensive because someone must be paid to sell insurance, and that compensation — the *variance risk premium* — means a high VIX predicts higher forward returns for a reason that has nothing to do with contrarian psychology (Bollerslev, Tauchen & Zhou 2009). Section 5 shows this is not a footnote: it is the largest single confound in the entire contrarian claim.
- **Flows carry lag and survivorship.** The free flow series are late, revised, and reported for the funds that still exist. A flow extreme you can see in real time on a free feed is usually already stale.

The operational upshot: there is no "sentiment" in the singular. There are three measurement systems, each partly a mirror of the recent past, each contaminated in its own way. A serious

sentiment model states which family it is in, prices the bias, and — above all — does not average the three as if they were independent. That last temptation is the subject of Section 4, and the reason we first have to look hard at positioning.

3 Positioning data

Positioning is the discipline's answer to the softness of survey sentiment: instead of asking people what they feel, it counts what they hold. Three sources matter — the Commitments-of-Traders report, ETF and fund-flow proxies, and options positioning — and they improve in that order as real-time, hard-to-fake gauges.

3.1 The Commitments-of-Traders report, decoded

Every Friday the U.S. Commodity Futures Trading Commission publishes the *Commitments of Traders* (COT), a breakdown of open interest in futures markets by trader category as of the preceding Tuesday. It is free, it is deep (the legacy futures-only files run to 1986 for many contracts), and it is the most-abused sentiment dataset in retail trading.

There are two report formats, and confusing them is the first error. The *legacy* report splits open interest into *commercial* (hedgers who deal in the physical or have a business reason to be in the futures), *non-commercial* (large speculators — funds, CTAs), and *non-reportable* (small traders, a residual). The *disaggregated* report, introduced after the 2008 crisis in response to criticism that "commercial" was a grab-bag, splits further into *producer/merchant/processor/user*, *swap dealers*, *managed money*, and *other reportables*. For a financial future like the E-mini S&P 500 the disaggregated "managed money" category is the cleaner read on speculative positioning, but the legacy non-commercial series has the longer history, which is why Figure 1 uses it. The disaggregated categories, for reference:

Disaggregated category	Who it is	How to read it
Producer/Merchant/Processor/User	Physical-market hedgers	Structural hedging demand; noisy as sentiment
Swap Dealers	Dealers laying off	Often the intermediary, not a view OTC/swap risk
Managed Money	CTAs, hedge funds, commodity pools	The cleanest speculative read
Other Reportables	Large traders not otherwise classified	Mixed motives

Three mechanical facts govern any honest use of the report. First, *the reporting threshold*: only traders holding positions above a CFTC size threshold are individually reported, so the categories cover the large players and lump everyone else into a non-reportable residual — the COT is a census of the big, not of the market. Second, *the timing and its leakage trap*: the report reflects positions as of Tuesday's close but is released Friday afternoon, a three-day lag, and any backtest that aligns the Tuesday-dated position to a Tuesday-or-later price without waiting for the Friday release has used data that did not exist yet. Figure 1 and Section 4's composite handle this by sampling each report at the Friday release, never the as-of Tuesday. Third, *why percentiles beat levels*: raw net contracts are not comparable across a period in which open interest grew fivefold, and even normalized by open interest the series has no natural zero for "extreme" — so the disciplined read is the *percentile of the current normalized reading within its own history*, which is exactly the 36th-percentile figure reported below. A percentile is regime-aware in a way a raw level is not.

Figure 1 shows net non-commercial positioning in E-mini S&P 500 futures (CFTC code 13874A) from 1998 to 2026, normalized by total open interest so that two decades of contract growth do not swamp the signal. The normalization matters: raw net contracts are not comparable across a period in which open interest grew from a few hundred thousand to over two million. Expressed as a percentage of open interest, large speculators have been net *short* the E-mini for 56% of all weeks — a structural tilt, because non-commercials are the counterparty to commercial hedgers who are net long index exposure for business reasons. The series has ranged from +50.3% net long (December 2000, at the dot-com top) to -50.8% net short (June 1998), with a mean of -1.1% and a standard deviation of 10.0 percentage points.

The **current reading**, computed live from the latest report in the cache (as-of 2026-09-08): large speculators are net short -3.7% of open interest — 76,036 net contracts against 2.07 million total open interest — which sits at the **36th percentile** of the full 1998-2026 history. In plain terms: mildly net short, but utterly unremarkable. This is the discipline the COT imposes and the folklore ignores — the honest output is a *percentile*, not a direction, and the honest percentile here says "nothing to see."

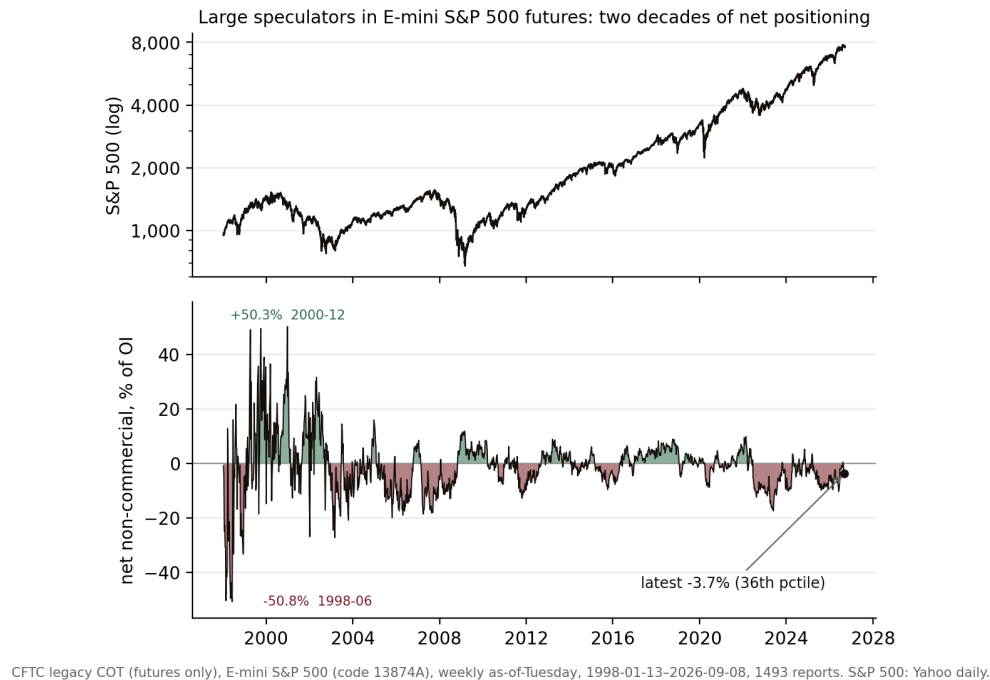


Figure 2. Large speculators in E-mini S&P 500 futures, 1998-2026. Top: the S&P 500 (log scale). Bottom: net non-commercial position (non-commercial longs minus shorts) as a percent of total open interest, weekly as-of-Tuesday, 1,493 reports; green above zero (net long), red below (net short). Extremes marked: +50.3% in December 2000 at the dot-com top, -50.8% in June 1998. The latest reading is -3.7% of open interest, the 36th percentile of the full history — mildly net short and unremarkable. Reproduce with `figures/fig_e3_cot.py`.

3.2 The commercials-versus-speculators frame and its pitfalls

The retail COT narrative is a morality play: commercials are the "smart money" (they know the physical market), speculators are the "dumb money" (trend-followers who will be caught offside at the turn), so *fade the speculators at extremes*. The frame is seductive, and mostly wrong, for four reasons a serious user must hold in mind.

First, *the categories are noisy labels, not identities*. A "commercial" swap dealer may be hedging a speculative structured-product book; a "managed money" fund may be running a genuine hedge. The CFTC itself introduced the disaggregated report because "commercial" had become a bucket of unlike motives.

Second, and decisively, *the predictive evidence is weak*. The most careful academic test of the "speculators are dumb money" hypothesis — Sanders, Irwin, and Merrin (2009), who examined whether COT positioning forecasts returns across agricultural futures — found essentially no systematic forecasting power; positioning-based signals did not beat simple benchmarks out-of-sample (Sanders, Irwin & Merrin 2009). The result does not prove COT is useless everywhere, but it shatters the confidence with which the "fade the specs" trade is usually asserted.

Third, *the frame confuses stock and flow*. An extreme *level* of speculative longs is often the residue of a strong trend; reading it as "about to reverse" is the COT version of calling a top because the market has gone up. The information, if any, is in the *change* and the *extremity relative to history*, not the sign.

Fourth, the counterexample the narrative never mentions: in Figure 1, speculators were near their *most net long* (+50%) in December 2000 — which the "fade the specs" story scores as a triumph, because the market then fell. But they were also deeply net *short* (−50%) in mid-1998, shortly before a sharp rally that would have crushed a "follow the smart-money commercials, who were long" position had it been sized to the extreme — and, more damning, for the long central band of the distribution (the 36th-percentile present included) the signal is simply noise. An indicator that is right at two cherry-picked dates and uninformative in between is not an edge; it is a pattern-matching device pointed at hindsight.

3.3 ETF and fund-flow proxies

Positioning can also be inferred from the plumbing of ETFs. When an equity ETF trades persistently above its net asset value, authorized participants create new shares, and *shares outstanding* rise; redemptions shrink them. Because shares-outstanding data is free (if coarse) and daily for large ETFs, the change in shares outstanding is a usable, if blunt, flow-of-positioning proxy — inflows into SPY/IVV/VOO as a read on equity demand, inflows into HYG/JNK as a read on credit appetite, GLD as a fear proxy. The caveats mirror Section 2.4: the signal is coincident-to-lagging, it is contaminated by arbitrage and tax-driven flows, and the *level* of assets tells you far less than the *rate of change* at extremes. The reading discipline mirrors the COT: it is the *rate of change relative to history*, not the level, that carries what little information exists, and the extreme in the tail rather than the drift in the middle. A useful cross-check is *divergence* — equity-ETF inflows continuing while credit-ETF inflows stall, or a broad-index inflow masking outflows from cyclical sectors — because a one-sided flow that is not confirmed across related assets is more likely noise than a genuine regime read. We do not build a figure here because the free shares-outstanding series are fragmented across issuers and awkward to assemble reproducibly; the honest statement is that this is a real but low-resolution gauge, best used to confirm a positioning read from a cleaner source, not to generate one.

3.4 Options positioning as sentiment — the cleanest real-time gauge

The argument of this section is that the single best real-time positioning read is not the COT and not fund flows but the *options market*, for three compounding reasons: it is continuous, it is expensive to fake (every position is posted premium and margin), and — uniquely — it feeds back into the underlying through *dealer hedging*, so options positioning is not merely a thermometer, it is part of the mechanism.

Three layers, increasing in information and in data cost:

- **Put/call ratios** (Section 2.3) are the crudest options-positioning read: aggregate demand for downside versus upside, free but discontinued in 2019. Useful as a coarse fear axis; too aggregated to locate *where* the positioning sits.
- **Skew** prices the asymmetry: persistently steep index skew reflects structural institutional demand for downside protection. As a *sentiment* signal it is muddled by that structural demand, but *changes* in skew — a sudden steepening, or an unusual bid for upside calls — are a cleaner read on shifting fear or greed.
- **Dealer positioning** is the deepest layer and the subject of the dealer-flows-and-GEX paper. Because option dealers hedge their inventory in the underlying, the *sign and size of aggregate dealer gamma* determine whether their hedging dampens or amplifies moves: long-gamma dealers sell rallies and buy dips (stabilizing), short-gamma dealers do the opposite (destabilizing). Estimating dealer gamma requires an options-chain feed and a set of assumptions about who is long what, which is why the best real-time versions are proprietary; the *concept*, however, is public and central. The informational content of options *order flow* — that option volume carries information about future stock prices, and that this is strongest where informed traders prefer the leverage of options — is documented in Pan and Poteshman (2006), whose put/call *volume* signal (built from open-buy volume, not raw volume) predicted next-day returns in their sample (Pan & Poteshman 2006). This is the empirical anchor for treating options flow as *informed* positioning rather than mere sentiment.

The mechanism that makes dealer positioning special — and that connects this section to the dealer-flows-and-GEX paper — is worth stating precisely, because it is what turns options positioning from a passive gauge into an active force. A dealer who is *short gamma* (net short options) has a hedge that must be adjusted in the *same* direction as the market: as price falls their position gets longer delta, so they sell to re-hedge, pushing price further down; as price rises they buy, pushing it up. Short-gamma dealers therefore *amplify* moves. A *long-gamma* dealer does the reverse, selling strength and buying weakness, *damping* moves. Two second-order Greeks extend the same logic across time and volatility: *charm* (the drift of delta as expiry approaches) drives predictable hedging flows into big expirations, and *vanna* (the change of delta as implied volatility moves) links the volatility surface to spot hedging, so a change in fear mechanically produces a hedging flow in the underlying. The Pan and Poteshman (2006) signal is the empirical complement to this structural story: rather than raw volume, they isolate *opening buys* of puts versus calls — trades that *initiate* a directional bet — and it is that informed subset, not aggregate volume, that predicts returns. The practical lesson is that "options positioning" is two different

things wearing one name: a *sentiment thermometer* (put/call, skew) and a *hedging-flow engine* (dealer gamma), and only the second feeds back into price.

The honest limit, and the reason this subsection ends in a caveat rather than a figure: the cleanest versions of options positioning — real-time dealer gamma, informed-flow decompositions — sit behind exactly the proprietary feeds this library refuses to publish from. What we can and do publish is the coarse free layer (put/call, Figure 2) and the *framework* for the rest, with the dealer-flows-and-GEX and flow-landscape mechanics as the load-bearing references. The claim "options positioning is the cleanest real-time sentiment gauge" is therefore well-supported in principle and only partially reproducible on free data — and we say so.

3.5 The current cross-section

It is instructive to read the free, measurable families *right now*, as the figures compute them, because the result is a useful antidote to the permanent temptation to see extremes everywhere. The market-based gauge (the VIX, Figure 2) sits at 17.7 — the **50.8th percentile** of its entire 1990-2026 history, i.e. exactly average. The positioning gauge (net speculator COT, Figure 1) sits at -3.7% of open interest, the **36th percentile** — mildly net short, unremarkable. The survey family we deliberately decline to quote a live number for, because the free series (AAII) is the weakest and the useful ones are paywalled. So across the two families a reproducible study can honestly measure, the reading is: *nothing*. No fear extreme, no complacency extreme, no crowded positioning. This is the normal state of the world — sentiment is *unremarkable* the overwhelming majority of the time — and the single most common analytical error is manufacturing a signal out of a middling reading because a signal was wanted. The disciplined output of a sentiment scan is usually "no edge today," and a framework that cannot say that is not a framework, it is a horoscope.

4 Building composite sentiment indicators

If no single sentiment series is both clean and free, the natural move is to combine several — a *composite*, in the tradition of Baker and Wurgler's (2006) sentiment index, which pooled six proxies into one factor (Baker & Wurgler 2006). Combination is genuinely useful: it averages away idiosyncratic noise and can surface a common signal. It also hides a trap that this section is built to expose.

4.1 Normalizing and combining signals

Raw sentiment series live on incompatible scales: the VIX is a volatility in points, the put/call ratio is a dimensionless ratio near 0.65, net COT positioning is a percent of open interest. To combine them you must first put them on a common footing, and the standard tool is the *z-score*

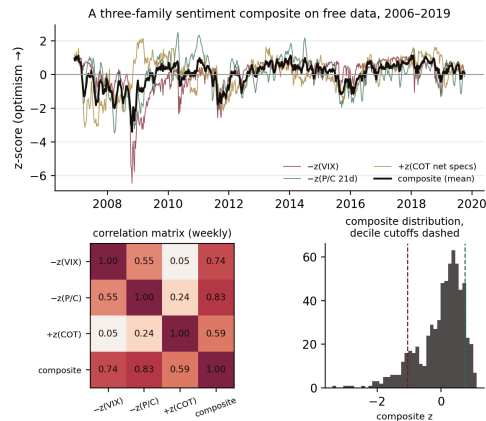
— subtract the mean, divide by the standard deviation, so each series becomes a unitless number of standard deviations from its own average. Signs are then aligned so that "high" means the same thing everywhere; we orient every component so that **high = optimism**: $-z(\text{VIX})$ (high VIX is fear, so we flip it), $-z(\text{put/call})$ (high put/call is fear, flipped), and $+z(\text{net specs})$ (speculators net long is, loosely, bullish positioning). The composite is the mean of the aligned z-scores.

Two subtleties separate an honest composite from a look-ahead artifact. First, *the mean and standard deviation used for the z-score must be knowable at the time* — using the full-sample mean to standardize a 2008 reading leaks the future into the past. Figure 3 is deliberately built the *dishonest* way, with full-window statistics, and labeled as descriptive only, because its job is to display *anatomy*, not to trade; the honest, expanding-window version is what Section 5's test uses. The z-score is not the only normalization, and the alternatives matter when series are fat-tailed, as sentiment series always are. *Rank (percentile) normalization* maps each reading to its position in the historical distribution, which is robust to outliers — a VIX of 80 and a VIX of 60 both map to "near the 100th percentile," so a single crisis does not dominate the standardized series the way it dominates a z-score. *Min-max* scaling to $[0, 1]$ is simple but hostage to the sample extremes. *Winsorizing* before z-scoring (clipping at, say, the 1st and 99th percentiles) is a middle path. There is no free lunch: the z-score is efficient if the series is roughly Gaussian and catastrophic if it is not, while the rank transform throws away magnitude to buy robustness. For sentiment, where the signal lives in the tails, the rank transform's compression of the tails is a real cost — it flattens exactly the extremes Section 5 cares about — which is why we keep the z-score here and pay the robustness price explicitly. The expanding-window discipline is the more important choice: standardize each observation using only the mean and standard deviation of data available *up to that point*, so the standardized value at time t could have been computed at time t . Full-window standardization (Figure 3) is a look-ahead and is used for description only; the tested composite in Section 5 uses in-sample statistics frozen before the out-of-sample half.

Second, *equal weighting is a choice, not a default* — and it is the wrong default when the components are correlated, which brings us to the exhibit.

4.2 A reproducible composite on free data — and its correlation matrix

Figure 3 assembles a three-family composite on free data, weekly on CFTC report dates (sampled at the Friday release so every input is knowable), over 2006-12 to 2019-10 — the only window in which all three series coexist, since the free put/call file dies in 2019. The top panel plots the three aligned components and their mean; the composite swings from deep pessimism in 2008-09 to optimism in the mid-2010s, as it should. The lower-left panel is the point of the whole section: **the correlation matrix of the components**.



Weekly on CFTC report dates sampled at the Friday release, 2006-12-01-2019-10-04, n=671. VIX & equity P/C: CBOE; COT: CFTC legacy (13874A). Full-window z-scores: descriptive only (look-ahead); the tested version is Figure 4.

Figure 3. A three-family sentiment composite on free data, 2006-2019 (671 weekly observations). Top: the three sign-aligned components — inverted VIX z-score, inverted put/call z-score, and net-speculator z-score — with their equal-weighted mean (the composite). Lower left: the component correlation matrix; the two market-based components correlate +0.55, while the COT component is nearly orthogonal to the VIX (+0.05). Lower right: the composite distribution with its 10th/90th-percentile decile cutoffs dashed. Full-window z-scores make this descriptive only (look-ahead); the tested version is Figure 4. Reproduce with `figures/fig_e3_composite.py`.

The matrix, in numbers:

	$-z(\text{VIX})$	$-z(\text{P/C})$	$+z(\text{COT})$	composite
$-z(\text{VIX})$	+1.00	+0.55	+0.05	+0.74
$-z(\text{P/C})$	+0.55	+1.00	+0.24	+0.83
$+z(\text{COT})$	+0.05	+0.24	+1.00	+0.59
composite	+0.74	+0.83	+0.59	+1.00

4.3 The multicollinearity problem

Read the off-diagonal. The two *market-based* components — inverted VIX and inverted put/call — correlate **+0.55**: they are two measurements of the same fear axis, exactly as Figure 2's +0.55 overlap correlation warned. The *positioning* component (COT) is nearly *orthogonal* to the VIX (+0.05) and only weakly related to put/call (+0.24): it carries genuinely different information. So an equal-weighted mean of the three does something statistically incoherent — it puts *two-thirds* of the weight on the fear axis (VIX and put/call) and *one-third* on positioning, then calls it a balanced composite. The composite's own loadings confirm it: it correlates +0.74 with the VIX component and +0.83 with the put/call component, but only +0.59 with COT. The "three-family" composite is, to first order, a fear index with a positioning garnish.

This is the *multicollinearity problem*, and it is the quiet killer of homebrewed sentiment gauges. When you average five "different" indicators that are really one factor measured five ways, you have not diversified — you have levered the common factor and fooled yourself into thinking a single noisy signal is a robust consensus. The fixes are all imperfect: drop redundant series (keep one market-based gauge, not three), weight by the inverse of the correlation (down-weight the redundant cluster), or extract the first *principal component* and treat *that* as the composite, letting the data decide the weights. Baker and Wurgler (2006) took the principal-component route precisely to avoid arbitrary weights. The lesson for the practitioner is blunt: *before you average sentiment indicators, run their correlation matrix*. If the off-diagonals are large, you own one indicator, not many — and you should test it as one. Which is exactly what Section 5 does.

5 The contrarian claim, tested honestly

The most durable slogan in sentiment trading is that *extremes reverse*: when everyone is fearful, buy; when everyone is greedy, sell. It is attributed to Buffett, printed on mugs, and almost never tested with a pre-registered protocol and an out-of-sample split. This section does exactly that, on free data, and reports whatever it finds.

5.1 The pre-registration

Following the discipline of the backtesting paper, the specification was frozen before the out-of-sample half was examined. It is reproduced here verbatim from the figure script's docstring.

```

# Pre-registration – does extreme sentiment reverse?

## Population & horizon
- Forward outcome: the 21-trading-day S&P 500 return following a reading.
- Series A (single cleanest): VIX close, CBOE, 1990-2026, daily.
- Series B (composite): the three-family composite of Section 4, 2006-2019, weekly.

## Confirmatory hypothesis (one, directional)
- H0: mean forward 21-day SPX return is EQUAL in the extreme-fear decile and
  the extreme-complacency decile.
- H1 (contrarian): fear-decile forward return > complacency-decile forward return.

## Metric
- Difference of mean forward returns, fear minus complacency.
- 95% CI via moving-block bootstrap (block = the 21-day horizon), which respects
  the overlap in forward returns; naive i.i.d. errors would be far too tight.

## In-sample / out-of-sample split (fixed dates; cutoffs frozen in-sample)
- Series A: in-sample <= 2008-12-31; OOS >= 2009-01-01.
- Series B: in-sample <= 2013-06-30; OOS >= 2013-07-01.
- Decile cutoffs computed on the in-sample half ONLY and applied, untouched, OOS.

## Decision rule
- Support H1 iff the difference is > 0 IN-SAMPLE and the SIGN HOLDS out-of-sample.

## Direction check
- "Fear" is the state the contrarian thesis says should earn HIGH forward returns:
  the TOP VIX decile (high VIX) and the BOTTOM composite decile (low optimism).

```

Two design choices deserve a word. The horizon is 21 trading days (about a month), long enough for a reversal to express itself and short enough that overlapping windows do not swallow the entire sample. And the confidence intervals are *moving-block bootstraps* with the block length set to the return horizon, because 21-day forward returns measured daily overlap by construction; a naïve standard error would treat 21 near-identical observations as independent and overstate significance by a factor of several. We also report an *overlap-adjusted effective n* ($\approx n$ divided by the horizon) so the minimum-cell discipline of Section 8.3 can be checked directly.

5.2 The single cleanest series: VIX deciles

Panel A of Figure 4 sorts every VIX close into deciles using in-sample (≤ 2008) cutoffs — a low-fear cutoff of 11.98 and a high-fear cutoff of 28.60 — and measures the forward 21-day S&P 500 return. The full-sample mean forward return is +0.81%, the ordinary monthly equity drift. Against that baseline:

Series A: VIX	In-sample ($\leq 2008-12-31$)	Out-of-sample ($\geq 2009-01-01$)
Extreme fear (top decile, VIX ≥ 28.6)	+1.58% 95% CI [-0.58, +3.77] n=479, n_eff=23	+3.29% 95% CI [+1.47, +5.50] n=434, n_eff=21
Middle 80%	+0.37% [-0.09, +0.86] n=3823	+0.90% [+0.42, +1.36] n=3691
Extreme complacency (bottom decile, VIX ≤ 12.0)	+0.55% [-0.02, +1.08] n=485, n_eff=23	+1.00% [+0.20, +1.72] n=308, n_eff=15
Fear – complacency	+1.02% [-1.14, +3.28] (CI includes 0)	+2.30% [+0.34, +4.65] (CI excludes 0)

The pre-registered decision: in-sample the fear-minus-complacency difference is positive (+1.02%), so the sign condition is met; out-of-sample the sign holds *and strengthens* to +2.30% with a bootstrap CI that excludes zero. **By the frozen rule, H1 is supported for the VIX.** Buying the extreme-fear decile earned about 2.3 percentage points more over the following month than buying extreme complacency, out-of-sample, over 2009-2026.

Now the honesty that separates this from a listicle. Three caveats, each load-bearing:

First, the effect was not significant in-sample. The in-sample difference's CI includes zero; only out-of-sample did it clear the bar. This is the *opposite* of the usual overfitting signature (in-sample strong, OOS weak), and it means the result is not a mined in-sample artifact — but it also means the "edge" is concentrated in the post-2009 regime, which was dominated by two enormous fear-then-rebound episodes (2020 and 2022). An indicator whose payoff lives in two events is an indicator with an effective sample size far smaller than its n suggests.

Second, and most important, most of "buy the fear" is not sentiment at all — it is the variance risk premium. A high VIX is a high *price* for index insurance, and Bollerslev, Tauchen, and Zhou (2009) show that a high variance premium predicts higher forward equity returns as *compensation for risk*, not as a reward for contrarian cleverness. The fear decile's +3.29% is partly a risk premium you earned by holding through genuinely dangerous months, some of which felt exactly as bad as the VIX said. Attributing it to "the crowd was wrong" is a category error. Disentangling the two is possible in principle but hard in practice: the variance risk premium is estimable as implied variance (VIX^2) minus a forecast of realized variance, and one can regress forward returns on the *sentiment* residual after projecting out that premium — but the two are so collinear (fear and the premium spike together) that the decomposition is unstable on any sample short enough to be relevant. The honest position is not to claim a clean split but to *state the confound loudly*: at least part, and plausibly most, of the fear decile's excess return is risk compensation, and a strategy that harvests it is short volatility in disguise, with the tail risk that implies.

Third — the crucial failure — the complacency half of the contrarian claim did not hold. The slogan is symmetric: buy fear *and sell complacency*. But the extreme-complacency decile ($VIX \leq 12$) was followed by *perfectly ordinary positive returns* — +0.55% in-sample and +1.00% out-of-sample, both positive, both near or above the middle-80% baseline. Low volatility did not precede poor returns; it preceded more of the same calm, upward grind. This is the single most important and least advertised fact about contrarian sentiment: **complacency is not bearish**. Selling because "the VIX is too low and everyone is complacent" has been, historically, a way to miss the best part of low-volatility bull regimes.

5.3 The composite: underpowered and regime-fragile

Panel B runs the identical protocol on the Section 4 composite, and the result is a lesson in limits rather than a signal.

Series B: composite	In-sample ($\leq 2013-06-30$)	Out-of-sample ($\geq 2013-07-01$)
Extreme fear (bottom decile)	+1.24% [-2.78, +4.61] n=35, n_eff=7	n = 0 (untestable)
Middle 80%	+0.49% [-0.68, +1.53] n=274	+1.01% [+0.31, +1.65] n=255
Extreme complacency (top decile)	-0.72% [-3.28, +1.10] n=35, n_eff=7	+0.18% [-0.91, +1.16] n=72, n_eff=14
Fear – complacency	+1.95% [-2.42, +6.32] (CI includes 0)	(CI untestable)

Two facts sink the composite as a testable contrarian signal, and both are instructive. First, *it is underpowered even in-sample*: with 35 raw observations per decile and a 21-day overlap, the effective n is about **7 per cell** — below the minimum-cell threshold of Section 8.3. The in-sample difference (+1.95%) points the contrarian way but its CI spans from -2.4% to +6.3%; it is consistent with a large effect, no effect, or a moderate *anti*-contrarian effect. Nothing can be concluded.

Second — and this is the more subtle failure — *the out-of-sample fear decile is empty* ($n = 0$). The decile cutoffs were frozen on the 2006-2013 in-sample window, which *contained the 2008-2009 crisis*, so the "extreme fear" threshold is a crisis-grade level of pessimism. The out-of-sample window (2013-2019) was a calm bull market that **never once revisited that level of fear**. The test is therefore not "the composite failed"; it is *"the composite could not be tested out-of-sample, because the state it defines as extreme never recurred."* This is regime-dependence in its purest form: an indicator calibrated on a crisis is silent in a calm, and an indicator calibrated on a

calm will scream falsely in a crisis. The honest verdict on the composite is a null of the most useful kind — not "no effect" but "insufficient evidence, by construction."



Figure 4. Does extreme sentiment reverse? Pre-registered decile test; forward 21-day S&P 500 return by sentiment decile, in-sample (solid) versus out-of-sample (hatched), 95% moving-block-bootstrap CIs. Panel A: VIX, 1990-2026. The extreme-fear decile (high VIX) earns +3.3% out-of-sample versus +1.0% for extreme complacency, a +2.3-point difference whose CI excludes zero — the contrarian claim holds for fear. Note that extreme complacency was still followed by ordinary positive returns; the "sell complacency" half fails. Panel B: the three-family composite, 2006-2019, is underpowered (effective n about 7 per decile in-sample) and its out-of-sample extreme-fear decile is empty (n=0), because the crisis-grade fear threshold frozen in-sample never recurred in the 2013-2019 calm. Reproduce with [figures/fig_e3_contrarian.py](#).

5.4 Sentiment versus momentum — when the crowd is right

The contrarian frame has a mirror image that is at least as well-evidenced: *momentum*. Prices that have risen tend to keep rising over 3-12 months; the crowd, in aggregate and most of the time, is *right* about direction, and fading it is a slow way to lose money. Barberis, Shleifer, and Vishny (1998) built the canonical behavioral model that reconciles the two — investor psychology (conservatism, representativeness) produces *underreaction* at short horizons (momentum: the crowd is right and not yet crowded enough) and *overreaction* at long horizons (reversal: the crowd is finally wrong) (Barberis, Shleifer & Vishny 1998). The two effects are not contradictory; they live at different horizons.

This is the resolution of the paper's central tension, and it is uncomfortable for the contrarian temperament. *Sentiment extremes are a long-horizon, low-frequency, valuation-anchored signal; momentum is a medium-horizon, higher-frequency, price-anchored signal*. Most of the time — in the wide middle of the sentiment distribution, which is where you spend 80% of your life — the crowd's direction is the better bet, and "everyone is bullish" is not a sell signal, it is a description of an uptrend. Contrarianism earns its keep only in the tails, only at the fear extreme (per Section 5.2), and only if you can survive the drawdown between "extreme" and "reversal," which in 2008

was another 45% and several months. The historical record is unambiguous on the failure mode: *fading a trend because sentiment is "too" bullish* has been, across 1995-1999, 2013-2019, and 2020-2021, one of the most reliable ways to underperform. The crowd is dangerous at the extremes and correct in the middle, and the hard skill is telling which regime you are in *before* the reversal, not after.

There is a deeper reason the two effects coexist rather than cancel, and it is worth making explicit because it dissolves a lot of confused argument. Momentum and reversal are not competing theories of the same horizon; they are the *same* behavioral machinery observed at two horizons. Underreaction to news — because people update their beliefs too slowly (conservatism) — produces a price that drifts toward fair value over weeks to months, which *is* momentum, and during that drift the crowd's direction is correct. Overreaction — because people extrapolate a run of good news into a trend that is not there (representativeness) — eventually pushes price past fair value, and the correction of that overshoot *is* long-horizon reversal, which is where contrarianism finally pays. The practical difficulty is that you cannot observe fair value, so you cannot see directly whether you are in the drift phase (follow) or the overshoot phase (fade); you can only estimate it from *how extreme and how crowded* the sentiment has become, which is noisy and lagging. This is why disciplined contrarianism is rare and disciplined trend-following is common: the trend gives a clean, price-based entry signal, while the reversal requires a judgment about an unobservable, made precisely when the social pressure to conform is highest. The empirical record of Section 5.2 fits this exactly — the payoff to fading lived only in the deepest fear tail, and even there it was concentrated in a handful of episodes, while the wide middle rewarded going with the tape.

6 News and text sentiment

The three families of Section 2 all reduce a market to a scalar mood. A richer, and now cheaper, source is *text*: news headlines, earnings-call transcripts, regulatory filings, social media. If sentiment is what people feel, text is where they say it in their own words, and natural-language processing (NLP) turns those words into numbers. This section surveys the method, dwells on the leakage traps that are *specific* to text, and describes a minimal reproducible pipeline — without any live scraping, which would make the paper unreproducible the day a website changes.

6.1 From headlines to a score

There are two families of text-sentiment method, and the choice is a genuine fork.

The *dictionary* (lexicon) approach counts words against a pre-built list of positive and negative terms. Its landmark in finance is Loughran and McDonald (2011), who showed that general-purpose sentiment dictionaries are actively *misleading* on financial text: words like "liability,"

"tax," "cost," "capital," and "board" are negative in a Harvard psychosocial dictionary but neutral or positive in a 10-K, so a naïve count mislabels ordinary disclosure as bad news. They built a finance-specific dictionary that corrected the error, and their word lists are freely available and became the field standard (Loughran & McDonald 2011). The dictionary method is transparent, fast, and reproducible — every score can be traced to specific words — which is exactly why it remains the right default for a paper that values reproducibility over headline accuracy. A concrete example makes the Loughran-McDonald point vivid: a company that writes "our tax liability increased as the board approved additional capital expenditure to address vice-president turnover" is making a neutral operational disclosure, but a general-purpose psychosocial dictionary flags *liability*, *tax*, *capital*, *vice*, and possibly *board* as negative — five false negatives in one anodyne sentence. Loughran and McDonald found that roughly three-quarters of the negative-word occurrences in 10-Ks under the general Harvard dictionary came from words that are not negative in a financial context, which is enough to swamp any genuine signal. Their fix — a hand-built finance dictionary — is not more sophisticated, it is more *correct about the domain*, and that is the general lesson: in text sentiment, domain-appropriate simple methods beat domain-blind complex ones.

The *model* approach learns sentiment from labeled examples: classical machine-learning classifiers on bag-of-words features, or modern transformer models (FinBERT and its descendants) fine-tuned on financial text. Models capture context and negation that a dictionary misses ("not bad" is positive; a lexicon counts "bad"), and they are more accurate on held-out text. They are also opaque, heavier to run, and — the point that matters for this library — *prone to hidden look-ahead through their training data*, discussed next.

The foundational empirical result that text moves markets is Tetlock (2007), who quantified the tone of the Wall Street Journal's daily "Abreast of the Market" column and found that high media pessimism predicted downward pressure on prices followed by a reversion to fundamentals, and that unusually high or low pessimism predicted higher trading volume (Tetlock 2007). Da, Engelberg, and Gao (2015) extended the idea to search behavior, building a "FEARS" index from the volume of Google searches for terms like "recession" and "bankruptcy" and showing it correlated with volatility and short-horizon reversals (Da, Engelberg & Gao 2015). Antweiler and Frank (2004), studying internet message boards, delivered the essential sobering counterweight: message-board "talk" carried a little information about volatility but was mostly noise, and the naïve trade did not work (Antweiler & Frank 2004). The literature's honest consensus is that text sentiment is *real but small and fragile* — a theme this whole paper keeps rediscovering.

6.2 The look-ahead and leakage traps specific to text

Text data has failure modes that price data does not, and they are subtle enough that published papers have fallen into them.

- **Timestamp leakage.** A news article's *publication* time is not its *content* time. Stories are updated, back-dated, and re-timestamped; a headline you read as of 09:00 may have been revised from a very different 08:00 version. Using the final revised text with the original timestamp leaks the future. The discipline is to freeze the *first* version at the *first* timestamp, which most free archives do not preserve — a reproducibility hazard, not a solvable bug.
- **Model-training look-ahead.** A transformer like FinBERT was trained on a corpus that *includes the period you are backtesting*. When you score a 2015 headline with a 2020-trained model, the model has, in a diffuse but real sense, already seen how 2015 turned out — the labels it learned encode outcomes. This is a uniquely textual, uniquely modern leak, and it silently invalidates most "our NLP signal predicts returns" backtests that use an off-the-shelf pretrained model over historical text. A dictionary frozen before the test window (Loughran-McDonald's lists predate most test samples) is *immune* to this, which is a second reason to prefer it for honest backtests.
- **Survivorship and selection in the corpus.** Free news archives over-represent large, surviving outlets and delisted-company coverage is patchy; the text you can download is not the text a trader saw in real time.
- **The publication-lag / anticipation confound.** Prices often move *before* the news is published, because the news reports what the market already knew. A text signal that "predicts" returns may simply be reading a slow news cycle that trails price — Tetlock's own finding of *reversion* after pessimism is consistent with news lagging fundamentals, not leading them.

6.3 A minimal reproducible pipeline (illustrative)

The following pipeline is described, not run live, precisely so that it stays reproducible and leak-free. It is deliberately a *dictionary* pipeline.

1. **Corpus (frozen).** Take a fixed, dated, redistributable text set — for example, a public archive of dated financial headlines saved once to `figures/data/` as a CSV with an immutable publication date per row. No live scraping: the corpus is a cached artifact, versioned like any other data.
2. **Preprocess.** Lowercase, strip punctuation, tokenize on whitespace. Do *not* stem aggressively; Loughran-McDonald's lists are word-form specific.
3. **Score.** For each document, $\text{tone} = (\text{n_positive} - \text{n_negative}) / \text{n_words}$, using the Loughran-McDonald finance word lists (public, frozen). Aggregate to a daily mean tone.
4. **Align without leakage.** Attach each day's tone to the *next* session's return — decision strictly before outcome — and, critically, use only the publication date, never a revision date.

5. **Test, pre-registered.** Exactly as in Section 5: one directional hypothesis, an in-sample/out-of-sample split by fixed dates, an effect size with a block-bootstrap CI, and a decision rule frozen first. Report whatever it shows.

A word on what such a pipeline should be *expected* to find, so that a positive result is read with appropriate suspicion. The published effect sizes for text sentiment on returns are small and short-lived: Tetlock's media-pessimism effect was a modest downward pressure that reverted within days; the FEARS search-based effect operated at a one-to-two-day horizon and reversed; message-board tone barely moved anything. A leak-free dictionary pipeline on daily headlines should therefore, at best, recover a *small, short-horizon, reversing* signal — and if it produces a large, persistent, monotone edge, the correct inference is not "we found alpha" but "we have a leak," most likely a timestamp or model-training look-ahead from Section 6.2. The evaluation discipline is the same as everywhere in this library: pre-register one directional hypothesis, split by fixed dates, report the effect size with a block-bootstrap CI, and check the effective n. A text signal that clears that bar out-of-sample is worth having; one demonstrated only in-sample, only on the dates it worked, is the Section 8.2 survivorship trap in its newest and most seductive form, because the sophistication of the NLP lends a false credibility to a result that was never tested.

The pipeline is intentionally boring. Every step is a place where a fancier choice (a pretrained model, an intraday timestamp, a live feed) would buy accuracy at the cost of reproducibility or leak-resistance, and for a paper whose thesis is that *most sentiment edges are survivorship*, the boring, auditable version is the honest one. We flag this section explicitly: it surveys a well-evidenced literature and specifies a clean method, but it does *not* contribute an own result, and it should not be read as one.

7 Crowding and fragility

Sentiment and positioning matter most not on average but at the breaking point, where an extreme in *who holds what* sets up a discontinuous move. This is the bridge to the flow-landscape paper (deleveraging cascades), and the organizing metaphor is combustion: *sentiment is the fuel — a crowded, one-sided position — and flows are the trigger — the forced transaction that ignites it*. Neither alone is sufficient. A crowded position that is never forced to unwind simply earns or bleeds slowly; a flow shock with no crowding is absorbed. The fragility lives in the conjunction.

7.1 Sentiment as fuel, flows as trigger

The mechanism has a specific shape. First, a period of calm and positive returns draws capital into a trade — long equities, short volatility, long a momentum factor — until the position is *crowded*: a large share of the available capital is on the same side, and the marginal buyer is nearly exhausted. Crowding is a *positioning* extreme, not a sentiment survey; it is measurable in

principle from flows, futures positioning, and options open interest, though rarely in real time on free data. Second, a *trigger* forces a subset of holders to transact against their will — a margin call, a risk-limit breach, a redemption, a mechanical rebalance. Third, because the position is crowded, the forced sellers meet no natural buyers and the price gaps; the gap triggers the next tranche of forced sellers (more margin calls, more stops, dealer hedging in the wrong direction); and the cascade feeds itself until the crowd is cleared. The flow-landscape paper formalizes the dealer-hedging amplifier — short-gamma dealers who must *sell* into a decline — that turns a stumble into a cascade.

Crowding itself is measurable, if imperfectly, and it is worth knowing the proxies even though none is clean on free data in real time. Futures positioning at an extreme percentile (the COT tail of Section 3) flags a one-sided speculative book. Concentrated options open interest at a strike or expiry flags where hedging flows will cluster. A collapsing cross-sectional dispersion of returns — everything moving together — flags that a single factor has become the whole market, the hallmark of a crowded factor trade. And a low, falling realized volatility under a rising asset price is the classic calm-before-the-storm signature that lets leverage and short-volatility positions accumulate. Each is a *fuel gauge*, not a *spark detector*; together they can tell you the room is full of gas, which is genuinely useful for sizing and hedging even though it says nothing about the match.

The counterexample that keeps this honest: *crowding is necessary but not sufficient, and it can persist for years without igniting*. Short-volatility strategies were crowded for most of 2016-2017 and paid handsomely; long-momentum was crowded through 2013-2019 and mostly kept working. "This position is crowded, therefore it will crash" is false as stated — it is the *trigger* that is missing, and the timing of the trigger is not predictable from the crowding alone. Crowding tells you the fuel is stacked; it does not tell you when the match arrives.

7.2 February 2018 — the short-volatility unwind

On 5 February 2018 the S&P 500 fell about 4% and the VIX more than doubled intraday, from the low teens to the high 30s — the largest one-day VIX spike in its history to that point. The proximate victims were *inverse and leveraged volatility ETPs*, above all the VelocityShares Daily Inverse VIX Short-Term ETN (XIV), which was designed to deliver the inverse of the daily return of short-dated VIX futures. Two years of eerily calm markets had made shorting volatility one of the most crowded trades in the market — the fuel. The trigger was mechanical: these products must buy VIX futures at the end of the day to rebalance their inverse exposure when volatility rises, so the very spike that hurt them forced them to buy more of what was spiking, into a market with no sellers. XIV lost about 96% of its value in a day and was liquidated.

February 2018 is the textbook illustration of the whole section: an extreme in *positioning* (crowded short-vol), a *mechanical flow trigger* (end-of-day rebalancing that was convex in exactly the wrong direction), and a *reflexive amplifier* (the hedging buy pressure fed the spike). Note what it was *not*: it was not a macro shock, not an earnings miss, not a recession. The fundamental news that day was negligible. The move was almost entirely a positioning-and-flow event — sentiment infrastructure eating itself — which is precisely why a purely fundamental analyst never saw it coming and a positioning-aware one could at least see the fuel, if not the exact spark.

7.3 2021 — the meme episode

The January 2021 episode in GameStop and a handful of other heavily-shorter small caps is the same physics with the sign flipped and the crowd relocated to retail. The fuel was a *crowded short*: professional short interest in GameStop exceeded its float, an extreme positioning imbalance. The trigger was a coordinated retail buying wave, organized on social media and amplified by brokers' zero-commission, gamified order flow, that forced shorts to cover into no supply — a *short squeeze* — while simultaneous heavy call-buying forced option dealers, who were short those calls, to buy stock to hedge (a *gamma squeeze*, the mechanism of the dealer-flows-and-GEX paper running in reverse). Price rose roughly twentyfold in weeks, brokers restricted buying under their own clearing-margin pressure (a second, opposite flow trigger), and the position eventually cleared with enormous losses on the short side and a wide dispersion of outcomes on the long side.

The meme episode adds two lessons the 2018 case does not. First, *text sentiment was, for once, causally central rather than epiphenomenal* — the coordination happened in measurable social-media text, connecting Section 6 to Section 7 directly, though the leakage traps of Section 6.2 make it far easier to narrate after the fact than to have traded before it. Second, it shows crowding extremes on *both* sides of the book at once (crowded shorts, then a crowded retail long), and the fragility came from their collision. The counterexample discipline applies here too: *most heavily-shorter stocks do not squeeze*. High short interest is a standing condition across dozens of names at any time; the squeeze needs the trigger, and the trigger — coordinated forced buying against a hard float constraint — is rare. Reading "high short interest" as "imminent squeeze" is the Section 3.2 error in a new costume.

7.4 The general shape

Abstracting from the two cases, fragility follows a repeatable template: a slow build of *one-sided crowding* under a calm regime, a *mechanical or forced trigger* that is convex against the crowd, and a *reflexive amplifier* — dealer hedging, ETP rebalancing, margin spirals — that makes the move feed itself. Sentiment measures tell you the fuel is accumulating (extreme positioning, low

volatility, complacent surveys); they do *not* time the trigger, and — the recurring humility of this paper — *the interval between a visible crowding extreme and its ignition has ranged from days to years, and is not forecastable from the crowding alone*. The practical use of sentiment and positioning data in this context is therefore not prediction but *preparation*: knowing that the fuel is stacked changes how you size, hedge, and set stops, even though it cannot tell you the date of the match.

This reframing — from prediction to preparation — is the single most useful takeaway of the section, and it generalizes the whole paper. A sentiment or positioning extreme is a statement about the *conditional distribution* of future outcomes, not about the *mean*. When the short-vol trade is crowded, the expected return may still be positive (the trade keeps paying until it does not), but the distribution has grown a fat left tail: the same conditions that let the crowd accumulate are the conditions under which a trigger produces a cascade. The correct response to "the fuel is stacked" is therefore not to flip directional — the Section 5 lesson that complacency is not bearish still binds — but to treat convexity as underpriced: buy cheaper tail protection, reduce leverage, widen stops to avoid being the forced seller, and refuse to be the marginal crowded holder. The flow-landscape paper formalizes the cascade dynamics; the contribution of sentiment and positioning analysis is to tell you *when the system is primed for one*, which is a different and more honest claim than "a crash is coming."

8 Limits

Every preceding section has ended in a caveat; this one collects the limits into a standing discipline, because the failure mode of sentiment analysis is not that it is useless but that it *looks* useful in exactly the ways that do not survive contact with new data.

8.1 Noisy, revised, and regime-dependent

Sentiment data is *noisy*: the signal-to-noise ratio in the middle 80% of every series we examined is essentially zero, and what little signal exists lives in the tails, where by definition the sample is smallest. It is *revised*: survey series are re-based, put/call methodologies change (the CBOE altered its equity-volume definition in 2012, mid-sample in Figure 2), and flow data is restated. And it is *regime-dependent* in the strong sense demonstrated in Section 5.3 — an indicator calibrated on one regime can be literally untestable in the next, because the state it calls "extreme" does not recur. A sentiment threshold is not a law of nature; it is a summary statistic of the period you fit it on, and it inherits that period's regime. The VIX contrarian result of Section 5.2 is real *and* concentrated in two post-2009 fear events; report it that way, or misreport it.

8.2 The survivorship of "it worked" indicators

The sentiment-indicator landscape is a survivorship graveyard. For every put/call rule, AAI contrarian trigger, or COT extreme that is quoted because it "called" a turn, there are dozens that failed and are never mentioned, and the ones that are quoted were usually *selected after the fact* for having worked on the sample that made them famous. This is the multiple-testing problem the backtesting paper formalizes — the best-looking indicator among many tried is an order statistic, not a discovery — applied to sentiment specifically. Three tells that an "it worked" claim is survivorship: it is demonstrated on the *same* sample that motivated it (no out-of-sample split), it is shown only on the dates where it worked (no full-period base rate), and it has *no stated failure* (violating the rule that every causal claim names its exception). A sentiment indicator with a clean out-of-sample test and an honest list of the times it failed is worth more than ten that are merely famous.

8.3 Minimum-cell discipline ($n < 15$)

The Quant track imposes a hard floor: *a conditional statistic computed on fewer than fifteen effectively-independent observations is not reported as a finding*. Sentiment analysis violates this constantly, because conditioning on an *extreme* is conditioning on a *rare* state, and overlapping forward returns shrink the effective sample far below the nominal count. Section 5 made this concrete twice: the VIX fear decile had a healthy effective n of ~ 21 -23, so its result is reportable; the composite decile had an effective n of ~ 7 in-sample and *zero* out-of-sample, so its result is explicitly *not* reportable as a signal, only as a limit. The discipline is not pedantry — it is the difference between the two panels of Figure 4, one of which is a finding and one of which is a cautionary tale, and the effective- n column is what tells them apart. Before quoting any conditional sentiment statistic, compute the effective n after the overlap adjustment; if it is below fifteen, the honest output is "insufficient evidence," not a number.

8.4 Reflexivity — the deepest limit

The subtlest limit is that measuring sentiment can *change* it. A sentiment indicator that becomes widely followed is subject to a financial Goodhart's law: once "extreme put/call means buy" is common knowledge, traders anticipate the signal, front-run it, and arbitrage away the very reversal it was built to catch — so the published edge decays precisely because it was published. This is not hypothetical. Several classic contrarian rules that "worked" in the 1990s literature weakened markedly in the 2000s and 2010s, consistent with crowding of the signal itself, though disentangling decay-from-arbitrage from decay-from-regime is generally impossible on the available samples. Reflexivity also runs the other way: because dealer positioning *feeds back into price* (Section 3.4), a positioning gauge is not a passive thermometer but part of the thermostat, so the act of many traders acting on a gamma read can move the underlying and invalidate the read.

The deepest version of the limit is therefore epistemological: sentiment is a property of a system that *includes the observer and the observer's tools*, and any sufficiently popular sentiment signal is, in the limit, a signal about how many people are watching that signal. The defense is not a better indicator but a permanent discount: assume every sentiment edge is partly arbitrated, size for that, and re-test on genuinely out-of-sample data rather than trusting a number that was famous for working on the past.

The overarching limit, and the note this paper ends on: sentiment and positioning are *real, measurable, and mostly already in the price*. The tradable residue is small, asymmetric (fear pays, complacency does not), concentrated in tails and rare events, and easy to overstate through data snooping, survivorship, and overlap-inflated significance. Treated with that humility — a family named, a bias priced, a hypothesis pre-registered, an effect size with an honest CI, and an exception stated — sentiment is a genuine, if modest, input. Treated as a mood to be read off a chart, it is a mirror in which a trader sees whatever they already believed. This paper is a long argument for the first stance.

Evidence basis

Every substantive claim in this paper rests on one of three kinds of basis, distinguished here so a reader can tell at a glance which is which — **own reproducible computation** (recomputed from free data or a self-contained model, reproducible via the cited script), **literature** (peer-reviewed article or textbook, cited inline), and **practitioner consensus** (desk practice with no clean citation, treated as the weakest). Nothing here is called *validated* on external sources alone; the stronger word requires an own out-of-sample computation.

Claim / result	Basis
The VIX and the equity put/call ratio over their free histories correlate +0.55 (two windows on one fear axis), and the latest VIX (17.7) sits at its 50.8th historical percentile (Figure 2)	Own reproducible computation — <code>figures/fig_e3_putcall.py</code>
Net non-commercial COT positioning in E-mini S&P 500 futures (1998–2026), its historical extremes, and the current –3.7% of open interest at the 36th percentile (Figure 1)	Own reproducible computation — <code>figures/fig_e3_cot.py</code>
In a three-family composite the two market gauges correlate +0.55 while COT is near-orthogonal (+0.05), so equal weighting double-counts the fear axis (Figure 3)	Own reproducible computation — <code>figures/fig_e3_composite.py</code>
Pre-registered contrarian decile test — the VIX fear decile earns +3.29% forward out-of-sample and beats complacency by +2.30% (CI excludes zero), while the "sell complacency" half fails and the composite is untestable out-of-sample (Figure 4)	Own reproducible computation — <code>figures/fig_e3_contrarian.py</code> ; OOS split
The VIX as the market-based investor fear gauge	Literature — (Whaley 2000)
A high variance risk premium predicts higher forward equity returns as risk compensation — the main confound in "buy the fear"	Literature — (Bollerslev, Tauchen & Zhou 2009)
COT positioning has essentially no systematic return-forecasting power, gutting the "fade the speculators" folklore	Literature — (Sanders, Irwin & Merrin 2009)
Option-buy volume carries information about future stock prices — options flow as informed positioning	Literature — (Pan & Poteshman 2006)
Behavioral underreaction (momentum) and long-horizon overreaction (reversal) reconcile trend-following with contrarianism	Literature — (Barberis, Shleifer & Vishny 1998)
General sentiment dictionaries mislabel financial text; finance-specific word lists are the field standard	Literature — (Loughran & McDonald 2011)
Dealer gamma is the deepest positioning layer (short-gamma amplifies moves, long-gamma damps them), but its cleanest real-time forms require proprietary feeds	Practitioner consensus — not independently verified
Crowding is the fuel and a mechanical flow trigger is the spark (the Feb-2018 XIV unwind, the 2021 meme	Practitioner consensus — not independently verified

Claim / result	Basis
squeeze); the interval from crowding to ignition is not forecastable	

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All references above are publicly accessible: peer-reviewed journal articles available through libraries and DOI resolution, with two (Baker & Wurgler 2007; Sanders, Irwin & Merrin 2009) available open-access. No proprietary, course, or trading-academy material is cited, and no paywalled data vendor's series is used in any figure.